

Non-normality and the Limits of Data-Driven Decomposition for Stable Oscillators:

A Perspective from Gait Biomechanics

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Abstract

Limit cycles, or stable periodic orbits, are ubiquitous in biological systems, from cardiac rhythms and neural oscillations to the rhythmic coordination of locomotion. Recent advancements in data-driven methods such as Dynamic Mode Decomposition (DMD) and neural ordinary differential equations (ODEs) promise to extract mechanistic insight from time-series observations of these systems, yet in practice they encounter a constellation of difficulties that are rarely discussed together. We present a unified perspective on why decomposing limit-cycle dynamics from data is fundamentally harder than the analogous problem for fixed-point attractors. Three interacting mechanisms conspire against standard approaches: (i) measurement noise systematically biases identified linear operators, shrinking eigenvalues and corrupting mode shapes in ways that total-least-squares methods only partially remedy; (ii) the Floquet structure of limit cycles induces *intrinsic non-normality* in any single-operator approximation, so that eigenvalue decomposition fails to separate dynamical contributions even when the spectrum is perfectly estimated; and (iii) the phase-dependent geometry of stable isostable foliations—twisting Floquet eigendirections and varying contraction rates—means that no time-invariant linear model can faithfully represent the transverse dynamics without mode mixing. We develop diagnostic tools based on pseudospectral analysis and resolvent gain curves, and apply them to a Hankel-DMD pipeline for human locomotion data (299 gait trials across able-bodied controls and stroke survivors). We show that non-normality metrics derived from the DMD operator discriminate clinical populations with large effect sizes ($p < 0.001$), suggesting that the very structure that makes limit cycles hard to decompose also encodes clinically relevant information about gait stability. We discuss implications for neural ODE approaches, spectral submanifold methods, and piecewise-linear strategies, arguing that the community should embrace non-normality as a feature rather than a nuisance.

Keywords: Dynamic Mode Decomposition, limit cycles, non-normality, Floquet theory, pseudospectra, gait biomechanics, stroke rehabilitation, neural ODEs

1 Introduction

Rhythmic behavior is among the most fundamental modes of biological organization. The heartbeat, respiratory rhythm, circadian clocks, neural oscillations, and locomotor coordination are all examples of *limit cycles*—isolated periodic orbits that attract nearby trajectories [Strogatz, 2015, Guckenheimer & Holmes, 1983]. Unlike fixed points, limit cycles involve continuous motion around a closed orbit, with perturbations relaxing back to the cycle through transverse contracting directions described by *Floquet theory* [Floquet, 1883, Chicone, 2006]. Understanding the geometry of this relaxation—the Floquet multipliers and their associated eigendirections—is critical for predicting how biological oscillators respond to perturbation, for designing therapeutic interventions, and for building predictive models.

The past two decades have seen an explosion of data-driven methods for dynamical systems. Dynamic Mode Decomposition (DMD) [Schmid, 2010, Kutz et al., 2016] extracts a best-fit linear operator from time-series snapshots; its variants (extended DMD, Hankel-DMD, optimized DMD) have been applied to fluid flows, neuroscience, epidemiology, and robotics. Neural ordinary differential equations (neural ODEs) [Chen et al., 2018] offer a non-linear alternative, learning continuous-time dynamics from data through backpropagation. Spectral submanifold (SSM) methods [Haller & Ponsioen, 2016, Cenedese et al., 2022] seek to identify the lowest-dimensional invariant manifolds that capture the essential dynamics. All three families of methods have been motivated, benchmarked, and primarily validated on systems with *fixed-point attractors*—damped oscillations returning to equilibrium.

When these methods are applied to *limit cycles*, they encounter a qualitatively different landscape of difficulties that the fixed-point intuition does not prepare practitioners for. The purpose of this perspective is to lay out these difficulties in a unified framework, grounding the discussion in both theory and a concrete application to human gait data. Our central thesis is:

The combination of noise bias, intrinsic non-normality, and phase-dependent transverse geometry makes limit cycles fundamentally resistant to data-driven decomposition into separable, interpretable components—and this resistance is itself a meaningful dynamical signature that can be exploited clinically.

We organize the perspective as follows. Section 2 reviews the essential theory: Floquet analysis, DMD and its noise sensitivity, and the sources of non-normality in single-operator limit-cycle models. Section 3 describes our Hankel-DMD pipeline with stability enforcement, and the pseudospectral and resolvent-based diagnostic toolkit we have developed. Section 4 presents a controlled study on the Stuart–Landau oscillator, where ground-truth Floquet structure is known, demonstrating how non-normality emerges even from a nearly diagonal system. Section 5 applies the full pipeline to 299 gait trials from able-bodied controls and stroke survivors, showing that non-normality metrics discriminate clinical populations. Section 6 discusses why neural ODEs, piecewise-linear models, and spectral submanifold methods face related or compounded difficulties. Section 7 synthesizes the results and argues for a shift in how the community approaches data-driven limit-cycle modeling.

This work builds on Winner *et al.* [Winner et al., 2023], who used LSTMs to predict and find low-dimensional structure in human locomotion joint-angle data, demonstrating that

data-driven methods can capture gait dynamics but leaving open the question of *mechanistic interpretability*—what do the learned representations correspond to in dynamical systems terms?

2 Background: Why Limit Cycles Are Different

2.1 Floquet Theory and the Monodromy Matrix

Consider a dynamical system

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}), \quad \mathbf{x} \in \mathbb{R}^n, \quad (1)$$

possessing a stable limit cycle γ with period T . Linearizing about a point $\mathbf{x}_0 \in \gamma$ yields the variational equation

$$\dot{\boldsymbol{\xi}} = \mathbf{J}(\mathbf{x}_0(t)) \boldsymbol{\xi}, \quad (2)$$

where $\mathbf{J}(\mathbf{x}_0(t)) = D\mathbf{f}|_{\mathbf{x}_0(t)}$ is the *time-periodic* Jacobian. By Floquet’s theorem, the fundamental matrix solution $\Phi(t)$ can be factored as

$$\Phi(t) = P(t) e^{Bt}, \quad (3)$$

where $P(t) = P(t + T)$ is T -periodic and B is a constant matrix. The *monodromy matrix* $M = \Phi(T)$ maps perturbations through one full period; its eigenvalues μ_i are the *Floquet multipliers*, and the corresponding eigenvectors define the directions along which perturbations grow or decay.

One multiplier is always $\mu_1 = 1$ (the marginal direction tangent to the orbit). The remaining multipliers $|\mu_i| < 1$ for a stable limit cycle, and their associated *isostable directions* [Wilson & Ermentrout, 2016, Kvalheim & Revzen, 2021] define the transverse contracting geometry. Crucially:

- (i) The Floquet eigendirections are generally *not constant*—they rotate and stretch as one moves along γ , parameterized by the periodic matrix $P(t)$.
- (ii) The contraction rates can be *phase-dependent*: a perturbation may transiently grow in certain phases of the cycle before ultimately decaying.
- (iii) When multipliers are close in magnitude (small separation of timescales), the corresponding eigendirections can become nearly parallel, inducing geometric sensitivity.

These features have no analogue in fixed-point dynamics, where the eigenvectors of the Jacobian are constant and the contraction is uniform in time.

2.2 DMD as a Single-Operator Approximation

Standard DMD [Schmid, 2010] seeks a matrix $\tilde{A}$ such that $\mathbf{x}_{k+1} \approx \tilde{A}\mathbf{x}_k$. Given snapshot matrices $X = [\mathbf{x}_1 \cdots \mathbf{x}_{m-1}]$ and $Y = [\mathbf{x}_2 \cdots \mathbf{x}_m]$, the solution is

$$\tilde{A} = YX^+ \approx U_r^* Y V_r \Sigma_r^{-1}, \quad (4)$$

where $X = U\Sigma V^*$ is the SVD and r is the truncation rank.

For a system near a fixed point, $\tilde{A}$ approximates the matrix exponential $e^{J\Delta t}$, and the DMD eigenvalues and modes correspond directly to the eigenvalues and eigenvectors of the Jacobian J . The relationship is clean: the spectrum of $\tilde{A}$ carries the full dynamical information.

For a limit cycle, the situation is fundamentally different. The data $\mathbf{x}_k$ lie on or near a closed curve, and a single linear operator $\tilde{A}$ must simultaneously:

- (a) Rotate states around the orbit (the oscillatory component),
- (b) Contract perturbations transverse to the orbit (the Floquet decay), and
- (c) Approximate the *time-varying* Floquet structure with a *time-invariant* operator.

Requirement (c) is the crux of the difficulty. The time-varying Jacobian $\mathbf{J}(t)$ is being compressed into a constant matrix, and the resulting $\tilde{A}$ must use *non-normal mode mixing* to reproduce the net-stable, phase-dependent dynamics.

2.2.1 Hankel-DMD and Delay Embedding

To increase the rank and observability of the linear representation, one typically applies Takens-style delay embedding [Takens, 1981, Brunton et al., 2017], constructing a Hankel (block-row) matrix:

$$H = \begin{bmatrix} \mathbf{x}_1 & \mathbf{x}_2 & \cdots & \mathbf{x}_{m-\tau} \\ \mathbf{x}_2 & \mathbf{x}_3 & \cdots & \mathbf{x}_{m-\tau+1} \\ \vdots & & & \vdots \\ \mathbf{x}_{\tau+1} & \mathbf{x}_{\tau+2} & \cdots & \mathbf{x}_m \end{bmatrix} \in \mathbb{R}^{n(\tau+1) \times (m-\tau)}, \quad (5)$$

where τ is the number of delay lags and n is the ambient dimension. DMD is then applied to consecutive column pairs of H . This ‘‘Hankel-DMD’’ approach dramatically expands the state space, enabling the linear operator to represent richer dynamics—but it also amplifies the non-normality problem, as we will show, because the delay coordinates introduce additional geometric entanglement.

2.3 Noise Bias: A Universal Problem, Amplified by Oscillations

Consider the noisy observation model $\mathbf{y}_k = \mathbf{x}_k + \boldsymbol{\eta}_k$, where $\boldsymbol{\eta}_k$ is i.i.d. measurement noise. Standard DMD solves

$$\tilde{A} = \arg \min_A \|\mathbf{Y} - \mathbf{A}\mathbf{X}\|_F, \quad (6)$$

which is a *forward* least-squares problem: noise in $\mathbf{X}$ is treated as exact, while noise in $\mathbf{Y}$ is minimized. This asymmetry systematically biases the identified eigenvalues *toward the origin* (for continuous-time, toward $-\infty$ on the real axis), an effect well-known in the errors-in-variables literature [Fuller, 1987] and studied specifically for DMD by Dawson et al. [2016], Hemati et al. [2017].

For a fixed-point attractor, this bias makes estimated decay rates slightly too fast—a quantitative error, but not a qualitative one. For a limit cycle, the consequences are more severe:

- (i) **Spectral radius shrinkage:** The Floquet multiplier at $|\mu| = 1$ (the phase direction) is pulled inside the unit disk, making the orbit appear to spiral inward. The oscillation decays spuriously.
- (ii) **Frequency corruption:** Eigenvalue bias is generally not radially symmetric, so oscillation frequencies can shift.
- (iii) **Mode-shape distortion:** Since the bias depends on the signal-to-noise ratio of each mode, and oscillatory modes have high energy, the relative bias differs across modes, distorting the eigenvector structure.

Optimized DMD (OptDMD, Askham & Kutz 2018), total-least-squares DMD [Hemati et al., 2017], and forward-backward DMD [Dawson et al., 2016] partially address this by treating noise in both X and Y . However:

Neural ODEs [Chen et al., 2018] and other continuous-time neural network approaches are equivalent to a forward model in this respect: they learn $\dot{\mathbf{x}} = f_\theta(\mathbf{x})$ by minimizing forward prediction error, treating the state trajectory as ground truth. They cannot account for noise in the state observations themselves, and thus suffer from the same systematic bias.

This is a critical and underappreciated point. The community has largely treated neural ODEs as a “more flexible” alternative to DMD, but they inherit exactly the same noise-bias structure and additionally lack the spectral interpretability that makes DMD’s bias diagnosable.

2.4 Non-Normality: The Eigenvalue Decomposition Fails

Even if noise bias could be perfectly corrected, the eigenvalue decomposition of a limit-cycle DMD operator would *still* not decompose the dynamics into separable components. The reason is *non-normality*.

A matrix A is *normal* if $AA^* = A^*A$. Normal matrices have orthogonal eigenvectors and their spectral decomposition is well-conditioned. For non-normal matrices, eigenvectors can become nearly parallel, and the resolvent $(zI - A)^{-1}$ can be enormous even far from any eigenvalue. The degree of non-normality is measured by:

- The **eigenvector condition number** $\kappa(V)$, where V is the eigenvector matrix: $\kappa(V) = \|V\| \cdot \|V^{-1}\|$.
- The **Henrici departure from normality**: $\delta_H(A) = \sqrt{\|A\|_F^2 - \sum_i |\lambda_i|^2} / \|A\|_F$.
- The **Kreiss constant** $\mathcal{K}(A)$, bounding transient growth: $\sup_{n \geq 0} \|A^n\| \geq \mathcal{K}(A)$ [Trefethen & Embree, 2005].
- The **ε -pseudospectral radius** $\rho_\varepsilon(A) = \max\{|z| : z \in \sigma_\varepsilon(A)\}$.

2.4.1 Why Limit-Cycle DMD Operators Are Necessarily Non-Normal

Consider the simplest limit cycle: the Stuart–Landau oscillator

$$\dot{z} = (\sigma + i\omega)z - z|z|^2, \quad (7)$$

where $z \in \mathbb{C}$, $\sigma > 0$ controls the contraction rate, and ω is the natural frequency. On the limit cycle ($|z| = \sqrt{\sigma}$), the linearized transverse dynamics have Floquet multiplier $\mu_s = e^{-2\sigma T}$ with $T = 2\pi/\omega$.

Even for this analytically tractable system, a Hankel-DMD operator $\tilde{A}$ exhibits:

- $\kappa(V) \sim 10^5$ (compared to $\kappa(V) = 1$ for a normal matrix),
- Henrici number $\delta_H \approx 0.99$,
- Peak transient growth $\|\tilde{A}^n\| \sim 10^4 \times$ the asymptotic bound.

The mechanism is clear: the operator must simultaneously encode *rotation* (eigenvalues on the unit circle) and *contraction* (eigenvalues inside the unit circle), but the *exact cancellation* required for stable periodic orbits forces the eigenvectors into near-parallel configurations. The transient growth represents the operator’s internal mechanism for achieving the net rotation-plus-contraction: it amplifies certain combinations of modes before they are canceled by others.

This non-normality is **not an artifact of noise, finite data, or poor algorithm choice**—it is an **intrinsic consequence** of approximating a time-varying process with a time-invariant operator.

2.4.2 The Pseudospectral and Resolvent Perspective

The ε -pseudospectrum of A is defined as

$$\sigma_\varepsilon(A) = \{z \in \mathbb{C} : \sigma_{\min}(zI - A) < \varepsilon\} = \{z \in \mathbb{C} : \|(zI - A)^{-1}\| > \varepsilon^{-1}\}, \quad (8)$$

and reveals the true sensitivity of the dynamics. For our discrete-time setting (eigenvalues on or near the unit circle), we evaluate the **resolvent norm** at $z = re^{i\theta}$ with r slightly greater than 1:

$$G(f; r) = \|(re^{i\theta}I - \tilde{A})^{-1}\|, \quad f = \frac{\theta}{2\pi}f_s, \quad (9)$$

where f_s is the sampling frequency. This *resolvent gain curve* reveals which frequencies are most amplified by near-resonance with the non-normal operator—a forcing-response perspective that is natural for understanding how perturbations couple to the limit-cycle dynamics.

Key derived metrics include:

- **Stability margin:** $d(\tilde{A}) = \min_\theta \sigma_{\min}(e^{i\theta}I - \tilde{A})$, the smallest perturbation that destabilizes the system.
- **Kreiss constant:** $\mathcal{K}(\tilde{A}) = \sup_{|z|>1} (|z|-1)\|(zI - \tilde{A})^{-1}\|$, bounding worst-case transient growth.
- **Numerical abscissa:** $\omega(\tilde{A}) = \lambda_{\max}(\frac{1}{2}(\tilde{A} + \tilde{A}^*))$, controlling one-step energy growth.

3 Methods

3.1 Gait Data and Clinical Populations

We analyze a dataset of $N = 299$ overground walking trials from three populations:

- **AB** (Able-Bodied): 180 trials from 22 subjects, treadmill speeds 30–220 cm/s.
- **HF** (Higher-Functioning stroke): 65 trials from 11 subjects, speeds 40–127 cm/s.
- **LF** (Lower-Functioning stroke): 54 trials from 9 subjects, speeds 22–60 cm/s.

Each trial consists of $T = 1500$ time steps at $f_s = 100$ Hz (15 seconds) of $n = 6$ joint angles (bilateral hip, knee, ankle), preprocessed as described in Winner et al. [2023]. **[TODO: Confirm exact joint set and preprocessing.]**

3.2 Hankel-DMD Pipeline

For each trial, we construct a block-Hankel matrix (eq. (5)) with $\tau = 10$ delay lags, yielding a state dimension of $n_{\text{state}} = 6 \times 11 = 66$. We then:

1. Compute the economy SVD $H_X = U\Sigma V^*$, where H_X and H_Y are the lagged and shifted Hankel blocks.
2. Truncate at the noise floor: retain singular values $\sigma_i > 10^{-6}\sigma_1$.
3. Form the reduced operator $\tilde{A} = U_r^* H_Y V_r \Sigma_r^{-1}$.

3.2.1 Stability Enforcement

The raw DMD operator frequently has eigenvalues outside the unit disk (spectral radius $\rho > 1$), particularly for the stroke groups where noise levels are higher. **[TODO: Quantify: 68% of HF trials had $\rho > 1$.]** Since the underlying system is a stable limit cycle, these unstable eigenvalues are artifacts. We enforce stability by projecting each eigenvalue onto the unit disk:

$$\lambda_i \leftarrow \begin{cases} \lambda_i & \text{if } |\lambda_i| \leq 1, \\ \lambda_i/|\lambda_i| & \text{if } |\lambda_i| > 1, \end{cases} \quad (10)$$

and reconstructing $\tilde{A}_{\text{stable}} = W \text{diag}(\lambda_1, \dots, \lambda_r) W^{-1}$.

3.3 Non-Normality Metrics

For each trial’s stability-enforced operator $\tilde{A}$, we compute:

1. **Spectral metrics:** rank r , spectral radius ρ , leading singular value σ_1 .
2. **Non-normality metrics:** eigenvector condition number $\kappa(V)$, Henrici departure δ_H , peak transient growth $\max_n \|\tilde{A}^n\|$, Kreiss constant $\mathcal{K}$.

3. **Resolvent metrics (at $r = 1.01$ and $r = 1.05$):** peak resolvent gain (excluding DC), mean gain, gain at 1 Hz and 2 Hz (the fundamental and first harmonic of gait).
4. **Stability margin:** $d(\tilde{A})$ and the frequency at which it occurs.
5. **Normalized growth:** $\text{growth_per_rank} = \max_n \|\tilde{A}^n\|/r$, controlling for the rank confound.

Statistical comparisons use the Mann–Whitney U test (nonparametric, appropriate for unequal and non-normal sample sizes) with Cliff’s delta for effect size.

3.4 Neural ODE Experiments

[**TODO:** Describe the neural ODE training setup: architecture (MLP layers, hidden dim), training procedure (adjoint method), training/validation split, and the specific failure modes observed. Key points:

- Models converged to fitting the mean trajectory (the limit cycle) but failed to capture transverse dynamics.
- Forward simulation (generative mode) diverged rapidly—the models could not maintain a stable orbit.
- This is well-documented for periodic data in neural ODEs [Jordan & Dimanche, 2021, Linot et al., 2023].
- The fundamental issue: like standard DMD, neural ODEs treat $x(t)$ as noise-free and optimize a forward loss, so they cannot account for errors-in-variables.

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4 Results I: The Stuart–Landau Oscillator

As a controlled testbed, we first apply our pipeline to the Stuart–Landau oscillator (eq. (7)) with $\sigma = 0.1$ and $\omega = 2\pi$ (period $T = 1$ s). This system has a single nontrivial Floquet multiplier $\mu_s = e^{-0.2\pi} \approx 0.533$ and constant isostable directions (the radial direction in polar coordinates).

4.1 Ground-Truth Floquet Structure vs. DMD Representation

A Hankel-DMD operator with $\tau = 10$ and rank $r = 20$ yields:

- Spectral radius $\rho(\tilde{A}) = 0.99989$ (bias from $\rho = 1$).
- Eigenvector condition number $\kappa(V) \approx 1.5 \times 10^5$.
- Henrici departure $\delta_H \approx 0.99$.

[**TODO:** Figure 1: Stuart–Landau DMD dissection. (a) Eigenvalue spectrum (unit circle, eigenvalues clustered near it). (b) Transient growth curve $\|\tilde{A}^n\|$ vs n . (c) Pseudospectral contour plot showing ε -pseudospectra bulging far outside unit circle. (d) Resolvent gain curve at $r = 1.01$, showing resonance peaks.]

Figure 1: Non-normality anatomy of the Stuart–Landau DMD operator. Despite a spectral radius of 0.999, the operator exhibits transient amplification exceeding 10^4 and pseudospectra that extend well beyond the unit circle. This non-normality is intrinsic to the single-operator representation of limit-cycle dynamics.

- Peak transient growth $\max_n \|\tilde{A}^n\| \approx 1.4 \times 10^4$.

The key insight is that the Stuart–Landau system has the *simplest possible* Floquet structure—constant eigendirections, a single contraction rate—and yet the Hankel-DMD operator is profoundly non-normal. The non-normality arises not from the complexity of the underlying system but from the *representational mismatch*: a time-invariant linear operator must use eigenvector non-orthogonality to encode the interplay between oscillation and contraction.

4.2 Mechanism: Cancellation and Mode Mixing

The eigenvector matrix V of $\tilde{A}$ reveals the mechanism. Modes associated with eigenvalues near the unit circle (the “oscillatory” modes) have eigenvectors that are nearly parallel to those of the “decaying” modes. The operator achieves stable periodic orbits through *constructive interference on the limit cycle* (modes add coherently) and *destructive interference off the cycle* (transverse perturbations are canceled by near-parallel mode pairs with slightly different decay rates).

This cancellation mechanism means that:

1. Removing any single eigenvalue from the spectrum dramatically changes the predicted dynamics—modes are not separable.
2. The transient growth $\|\tilde{A}^n\|$ can vastly exceed the spectral-radius bound ρ^n , because the cancellation requires large intermediate amplitudes.
3. Small perturbations to $\tilde{A}$ (from noise) can qualitatively change which modes dominate, making the decomposition unstable.

5 Results II: Human Gait Data

5.1 Stability-Enforced DMD: Basic Metrics

Applying the pipeline to all 299 gait trials with stability enforcement yields the results in table 1.

The most striking result is the **growth per rank** metric: stroke survivors exhibit approximately $2.5\text{--}3\times$ the transient growth per mode compared to able-bodied controls ($p < 0.001$,

Table 1: Stability-enforced DMD metrics by group (median [IQR]).

Metric	AB ($n = 180$)	HF ($n = 65$)	LF ($n = 54$)
Rank r	45 [43–64]	37 [36–38]	36 [36–37]
$\kappa(V)$	73,000 [43K–121K]	159,000 [113K–263K]	136,000 [99K–214K]
δ_H (Henrici)	0.98 [0.98–0.99]	0.99 [0.93–1.00]	0.91 [0.89–0.97]
$\max_n \ \tilde{A}^n\ $	21,085 [7.5K–36K]	52,050 [37K–77K]	46,961 [29K–68K]
Growth/rank	475 [127–866]	1,375 [1,026–2,029]	1,294 [830–1,842]

[**TODO:** Figure 2: Strip/jitter plots of key metrics by group. (a) growth_per_rank: AB vs HF vs LF. (b) $\kappa(V)$ by group. (c) Kreiss constant by group. (d) growth_per_rank vs speed, colored by group.]

Figure 2: Non-normality metrics discriminate clinical populations. Growth per rank (a) shows the strongest separation ($p < 0.001$ for AB vs HF and AB vs LF). The effect persists after controlling for walking speed (d).

Cliff’s delta > 0.6 , “large” effect). This persists in speed-matched comparisons (30–60 cm/s), ruling out speed as a confound.

5.2 Pseudospectral Analysis

[**TODO:** Present pseudospectral results once cell 27 computation completes. Expected structure:

- Kreiss constant: $\text{HF} > \text{LF} > \text{AB}$.
- Resolvent gain at $r = 1.01$: stroke groups show larger peak gains, particularly near 1–2 Hz (gait fundamental and harmonic).
- Stability margin: AB has larger $d(\tilde{A})$ (more robust to perturbation).
- Pseudospectral contour plots for representative AB, HF, LF trials showing differential “bulging” beyond unit circle.

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5.3 Clinical Interpretation

The non-normality of the DMD operator is not merely a mathematical curiosity—it encodes how the gait system responds to perturbation. Higher growth-per-rank means that the limit-cycle dynamics rely on more delicate cancellations between modes to maintain stability. In physical terms:

- **Perturbation sensitivity:** A gait pattern with high non-normality can exhibit large transient excursions before returning to the periodic orbit, even if the orbit is formally stable (all Floquet multipliers inside the unit disk).

[**TODO:** Figure 3: Pseudospectral analysis. (a–c) ε -pseudospectral contours for representative AB, HF, LF trials. (d) Resolvent gain curves at $r = 1.01$ for the same three trials, with gait band (0.8–3 Hz) shaded. (e) Zoomed gait-band resolvent (0–5 Hz) at $r = 1.01$.]

Figure 3: Pseudospectral contours and resolvent gain curves reveal differential amplification structure across clinical populations. Stroke survivors’ DMD operators show larger ε -pseudospectra (more “leakage” beyond the unit circle) and higher resolvent gains at gait-relevant frequencies.

- **Robustness to model error:** Small changes in the operator (representing, e.g., fatigue, terrain changes, or neural control variability) have outsized effects on the predicted dynamics when the operator is highly non-normal.
- **Fall risk:** The transient growth magnitude directly bounds the worst-case perturbation amplification. If a stroke survivor’s gait control system has $3\times$ the transient growth of an able-bodied walker, their maximum excursion from the gait cycle in response to an identical perturbation is correspondingly larger.

[**TODO:** Discuss whether higher non-normality in stroke gait is a cause (maladaptive control strategy) or consequence (reduced neural bandwidth forcing the controller to operate closer to the stability boundary) of increased fall risk.]

6 Why Other Methods Face Similar or Worse Difficulties

6.1 Neural ODEs

Neural ODEs learn a vector field $\dot{\mathbf{x}} = f_\theta(\mathbf{x})$ by minimizing

$$\mathcal{L}(\theta) = \sum_k \|\mathbf{x}(t_k) - \hat{\mathbf{x}}(t_k; \theta)\|^2, \quad (11)$$

where $\hat{\mathbf{x}}(t_k; \theta)$ is the solution of the neural ODE integrated forward from an initial condition. This formulation has two fundamental problems for limit cycles:

- Noise-in-variables bias:** Like standard DMD, the loss eq. (11) treats the observed $\mathbf{x}(t_k)$ as exact. Noise in the state trajectory biases the learned dynamics in the same direction as for DMD (eigenvalue shrinkage / spurious damping), but the bias is harder to diagnose because the model is nonlinear and lacks a clean spectral decomposition.
- Error accumulation:** Forward integration compounds errors at each time step. For a limit cycle, even a small systematic bias causes the predicted orbit to spiral inward or outward, amplified by the integration horizon. Our experiments confirm this: neural ODEs trained on gait data converge to the mean trajectory but diverge rapidly in generative mode. [**TODO:** Include training curves and generative-mode divergence plots.]

These issues have been documented for periodic data by Jordan & Dimanche [2021] and Linot et al. [2023], who proposed stabilization tricks (e.g., augmented training losses that penalize divergence from the attractor). However, these fixes are application-specific and do not address the fundamental noise-bias problem.

6.2 Piecewise-Linear Models

An intuitive strategy for handling the time-varying Floquet structure is to partition the limit cycle into segments and fit a separate linear model in each:

$$\mathbf{x}_{k+1} \approx A_j \mathbf{x}_k \quad \text{for } \mathbf{x}_k \in \mathcal{R}_j, \quad (12)$$

where $\{\mathcal{R}_j\}$ partitions the state space. This approach faces compounding difficulties:

- (i) **Phase estimation:** Defining the boundaries $\{\mathcal{R}_j\}$ requires knowing the limit cycle and assigning a phase to each data point—itself a nontrivial task for noisy, high-dimensional data.
- (ii) **Boundary discontinuities:** At region boundaries, the switch between linear models introduces artificial discontinuities that accumulate during forward propagation.
- (iii) **Data fragmentation:** Each submodel has access to only a fraction of the data, increasing noise sensitivity and making the bias problem *worse* for each individual model.
- (iv) **No spectral coherence:** The eigenvalues of different A_j need not be related in any simple way, making it impossible to extract global Floquet multipliers without additional assumptions.

6.3 Spectral Submanifolds (SSMs)

Spectral submanifold methods [Haller & Ponsioen, 2016, Cenedese et al., 2022] identify the unique smoothest invariant manifolds tangent to spectral subspaces of the linearization. They offer a rigorous nonlinear extension of modal analysis—but with critical limitations for limit cycles:

- (i) **Fixed-point formulation:** The mathematical foundation of SSMs requires a fixed point with a well-defined spectrum. Limit cycles have a trivially zero eigenvalue (the phase direction), which violates the non-resonance conditions needed for SSM uniqueness.
- (ii) **Resonances:** The oscillatory eigenvalues of a limit cycle are inherently resonant with the periodic orbit, creating exactly the conditions that SSM theory assumes away.
- (iii) **Adiabatic SSMs:** The Haller group has developed “adiabatic SSMs” that essentially partition the limit cycle (as in the piecewise approach) and compute SSMs at each phase [Li et al., 2024]. As we noted for piecewise models, this reinvents the Floquet frame without resolving the fundamental difficulties of phase estimation and boundary management.

- (iv) **SINDy-based nonlinear Floquet:** Similarly, approaches using SINDy [Brunton et al., 2016] to learn a nonlinear Floquet decomposition [Kvalheim & Revzen, 2021] face the same challenge: they require accurate phase estimation and yield models whose coefficients vary along the orbit, deferring rather than resolving the decomposition problem.

7 Discussion

7.1 Embracing Non-Normality

The results of this study suggest a reframing: rather than viewing non-normality as a failure of the DMD approximation, we should view it as a *faithful encoding* of a genuine dynamical property. A limit cycle’s transverse relaxation is inherently non-normal when viewed through any single linear operator, and the degree of non-normality carries information about:

- (i) The separation of timescales between oscillation and contraction (small separation $\Rightarrow$ more non-normality).
- (ii) The phase-dependence and twisting of Floquet eigendirections (more complex geometry $\Rightarrow$ more non-normality in a time-invariant representation).
- (iii) The coupling between modes that the system relies on for stability (more cancellation $\Rightarrow$ higher transient growth).

The clinical result—that stroke survivors show $\sim 3\times$ the non-normality per mode—suggests that pathological gait is characterized by a more delicate balance between the oscillatory and contracting components of the dynamics. This may reflect:

- Reduced neural bandwidth for real-time gait adjustments, forcing the system to rely on fewer, more tightly coupled modes.
- Asymmetric limb dynamics (hemiparesis) that rotate the Floquet eigendirections more dramatically around the gait cycle.
- Compensatory strategies that trade off robustness for energy efficiency.

7.2 Implications for Data-Driven Modeling

Our analysis suggests several practical recommendations:

1. **Use total-least-squares or optimized DMD** to mitigate noise bias, and report the correction magnitude as a diagnostic.
2. **Report non-normality metrics** alongside eigenvalues whenever applying DMD to oscillatory data. Eigenvalues alone are misleading for non-normal operators.
3. **Interpret resolvent gains rather than eigenvalues** for frequency-domain analysis of limit-cycle systems.

4. **Be skeptical of forward-propagation accuracy** for any model (DMD, neural ODE, or otherwise) applied to limit cycles. The transient growth bound means that errors are amplified by $\max_n \|\hat{A}^n\|$ before settling to the spectral decay rate.
5. **Consider non-normality as a clinical biomarker** for movement disorders, complementing existing kinematic and kinetic measures.

7.3 Limitations and Future Directions

[TODO:

- Subject-level analysis (current analysis is trial-level).
- Extension to other biological oscillators (cardiac, neural).
- Development of non-normality-aware DMD variants.
- Connection to control-theoretic robustness margins for gait.
- OptDMD comparison.
- Comparison with phase-amplitude reduction methods.

]

8 Conclusion

Data-driven decomposition of limit-cycle dynamics is fundamentally harder than the analogous problem for fixed-point attractors. Noise bias, intrinsic non-normality, and phase-dependent Floquet geometry interact to make eigenvalue-based decomposition unreliable and forward propagation unstable. These difficulties are not specific to DMD—they affect neural ODEs, piecewise-linear models, and spectral submanifold methods in analogous ways. We have shown that the pseudospectral toolkit provides a principled diagnostic framework, and that the non-normality itself encodes clinically relevant information: stroke survivors’ gait dynamics exhibit significantly elevated transient growth, Kreiss constants, and resolvent gains compared to able-bodied controls. We argue that the community should shift from treating non-normality as a nuisance to exploiting it as a window into the mechanisms of biological oscillator stability.

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